

Computer Vision

Complex Computing Problem

CLO	Course Learning Outcome	GA	Complex Problem Attributes
CLO-3	Design different algorithms for spatial and frequency domain filtering, feature detection, structure from motion, motion estimation, etc.	GA-4	1. Depth of analysis required 2. Depth of knowledge required

Learning Objective

To enable students to utilize a pre-trained pose estimation model, analyze human body keypoints, compute joint angles, and design a rule-based system to classify human activities.

Scenario

You are required to design a pipeline that detects human poses, analyzes joint geometry, and classifies activities being performed. For your video source, you may use any one of the following options:

- **Webcam:** Record yourself or someone else performing two or more distinct activities (e.g., standing, sitting, raising arms, squatting).
- **Online Video:** Download any suitable clip from YouTube or any other platform that clearly shows a person performing distinct activities.
- **Pre-recorded Clip:** Use any existing video file you already have.

Note: The video should contain at least two distinct activities and the person should be clearly visible throughout.

Tasks

1. Pose Detection & Pre-processing

- Use a pre-trained model (e.g., MediaPipe Pose or OpenPose) to extract body keypoints from each frame.
- Apply a smoothing filter on the keypoint coordinates across frames to reduce jitter in the detections.
- Visualize the skeleton overlay on the original frames.

2. Joint Angle Computation & Tracking

- Select at least three meaningful joint angles (e.g., knee angle, elbow angle, hip angle) and compute them using the detected keypoints across all frames.
- Plot how these angles change over time and identify the frames where a transition between activities occurs.

3. Rule-Based Activity Classification

- Design a rule-based classifier using joint angle thresholds to distinguish between at least two activities.
- Report the classification result per frame and calculate the overall accuracy against your own manually labeled ground truth.

Deliverables

Item	Description
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Jupyter Notebook	Code implementation with visualizations and step-by-step comments for all three tasks (.ipynb or .py)
Short Report	1-2 page covering: angle computation approach, rule-based logic designed, accuracy results, observations and demo screen shots (.pdf)

Note.

- Clearly mention in your notebook which video source you used and provide the link or file name.
- Each student must submit independently.
- Commit your work on GitHub and paste the link of QOBE Portal